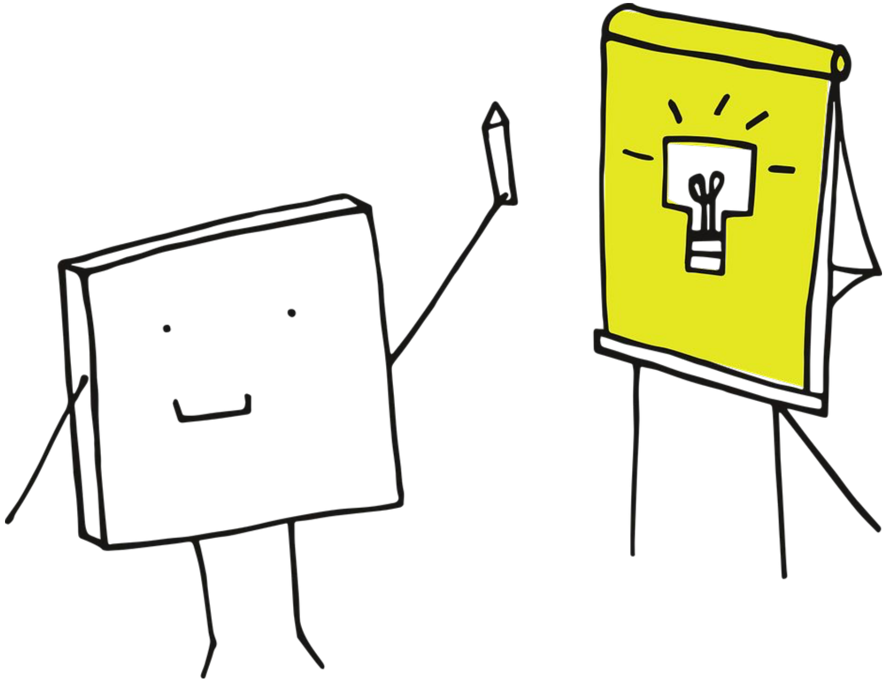
Lesson 15: Expressing knowledge



HS192 Introduction to Writing - 2
@ IITGN, 2025-26

Writing Project

Write down some important daily record of yours for next 10-14 days, between March 16 and 31. Your job is to make a table and visualize the information in the most effective way. The item can be:

- Your wake up times or bedtimes
- Your daily footsteps
- Your daily consumption of 3 meals + snack (e.g., ***Day 1:*** *lunch and dinner,* ***Day 2:*** *breakfast, snack, and dinner*)
- The hours spent for a favorite activity of yours
- The cups of chai you drink per day
- The bottles/cups of water you drink
- Duration of your phone conversation with a parent or relative.

Today's learning objective- 1

Identify ways of visually strengthening in-text contents.

Graphics for text

Visuals boost novice's understanding, to name a few...

(i) Photo/illustration
(ii) Diagram/flow chart/timeline
(iii) Graphs
(iv) Tables

These visuals must be chosen based on the effectiveness.

In our project, try to make two kinds of items out of (i)-(iv). In case of importing items from an external source, please cite them in the caption. Also, try to adjust the format to your document.

Nesting habits of the Tricoloured Munia *Lonchura malacca* in Veedur Reservoir in Viluppuram District, Tamil Nadu, India

M. Pandian

Pandian, M., 2022. Nesting habits of the Tricoloured Munia *Lonchura malacca* in Veedur Reservoir in Viluppuram District, Tamil Nadu, India. *Indian BIRDS* 18 (5): 135–141.
M. Pandian, No. F1901, Taisha, Natesan Nagar West, Virugambakkam, Chennai, Tamil Nadu 600092, India. E-mail: pandian.m14@gmail.com
Manuscript received on 23 April 2022.

Abstract

The nesting habits of the Tricoloured Munia *Lonchura malacca* were studied in the wetlands of Veedur Reservoir in Viluppuram District, Tamil Nadu. A total of 44 nests were enumerated on *Typha angustifolia*, *Prosopis juliflora*, *Vachellia nilotica*, and *Chrysogogon zizanioides*, and 365 birds counted in the study sites. The ball-shaped nests, with lateral entrances, took 4–8 days for construction, with participation of both male and female birds. 79.5% nests had entrances facing eastwards. Tricoloured Munia followed communal roosting and foraging, roosting on 12 plant species and foraging on 14. Incidents of Tricoloured Munia consuming filamentous green algae *Spirogyra irregularis*, and *Oedogonium crispum*, pig dung, and sand particles were noticed during the study period. Four pairs were observed mating on *Prosopis juliflora* trees. 11 nests were damaged due to harvesting of *T. angustifolia* reeds. No antagonistic interactions were found between Tricoloured Munia and congenerics, at foraging grounds and roosting sites. There were opportunistic sightings of House Crow *Corvus splendens*, Large-billed Crow *C. macrorhynchos*, Asian Koel *Eudynamus scolopaceus*, and Shikra *Accipiter badius* in the vicinity of nesting sites, but no predation was noticed, though there were incidents of Tricoloured Munia being chased by these birds during its foraging activity.

Introduction

The Tricoloured Munia *Lonchura malacca* is a small, finch-like, granivorous bird. It is native to southern Asia as well as parts of Indonesia and the Philippines. It has become naturalised in southern Europe and on several Pacific Islands (Payne 2020). In the Indian Subcontinent this species occurs from Gujarat to Sri Lanka (Ali & Ripley 1987; Grimmett et al. 2011; Rasmussen & Anderton 2012). The bird has been listed in various checklists from India: Rajasthan (Jamdar 1998; Sharma 2001; Bhatnagar et al. 2013), the Carnatic region (Jerdon 1840), Karur District (Salahudeen et al. 2013; Deepan et al. 2017), Coimbatore (Daniel 2017), and Viluppuram District in Tamil Nadu (https://ebird.org/species/trimun/IN-TN-VL). The Tricoloured Munia population is adapted to a wide range of habitats, such as open agricultural lands, woodlands, grasslands, and scrub lands. Its breeding has been studied in detail by Singh et al. (2001) in Odisha, where 81 nests were recorded on *Setaria plicata*, *Crinum asiaticum*, and *Cycas* spp.

In this paper, I document the nesting, roosting, and foraging habits of Tricoloured Munia in the wetlands of Veedur Reservoir in northern Tamil Nadu.

Materials and methods

Study area

The present study was carried out in the wetlands of Veedur Reservoir in Tindivanam Taluk, Viluppuram District (11.94°N, 79.49°E) of north-eastern Tamil Nadu (Fig. 1). The district covers 3,715 sq. km, with a human population of c.20,90,000 (https://www.viluppuram.nic.in/). Veedur Reservoir extends over an area of 7.7 sq. km with a storage capacity of 600 mcft water, and irrigates 1,295 Ha of arable lands. Agriculture is the primary occupation of the people here. The major crops of this taluk are Paddy *Oryza sativa*, Jowar *Sorghum bicolor*, Pearl Millet *Pennisetum glaucum*, Finger Millet *Eleusine coracana*, Sugarcane *Saccharum officinarum*, Groundnut *Arachis hypogaea*, and Green Gram *Vigna radiata*. Maximum and minimum temperatures in the district are 36°C and 20°C, respectively. The average annual rainfall is 1,060 mm (Fig. 1; Anonymous 2022).

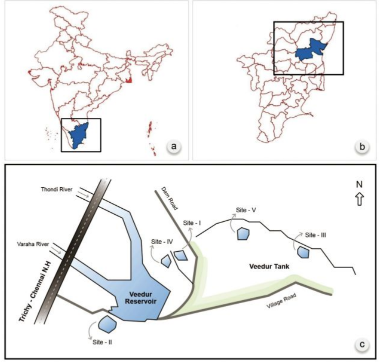


Fig. 1. A map of Veedur Wetlands in Viluppuram District, Tamil Nadu, India

Methods

With the help of two field assistants, I conducted a survey during the breeding season of the Tricoloured Munia, in the wetlands of Veedur Reservoir. This was done between 0545 and 1800 h, from the third week of December 2021 to the second week of March 2022. Variables, such as nesting sites, nest-supporting plants, nest materials, number of nests at each site, and the distance between nesting and foraging sites were quantified. The length and width of the nests, as well as the nests' entrances, and the height of nests from the water level below them were measured using a measurement tape after the birds had abandoned the nests. Orientations of nest entrances were determined using a compass in a smart phone. Roosting, foraging, mating, interactions with other associate bird species, and threats were observed through binoculars and from a reasonable distance (*c*.30 m). The number of birds was counted by following the total count method (Bibby et al. 2000) when the birds were day roosting and foraging. Bird censuses were carried out for two days, over four monthly sessions during every third week of a month, i.e., one day in Sites I & II and another day in Sites-III to V, and the maximum bird population size was taken as the total count. I identified the crop or grass species the bird used for roosting, and foraging. The filamentous green macroalgae eaten by the birds were identified using a compound microscope. A solitary nest in the grass species *Chrysopogon zizanioides*, was ignored during analysis. The measurement taken from the longest elliptical part of the outer surface of a nest was considered the nest length, the middle of the elliptical part as nest width, and similarly, the longest diameter of a nest's entrance was the entrance length, and the shortest diameter was considered the entrance width (Fig. 2). Based on the morphometric data of nest variables, mean and standard deviations were arrived at. Kruskal-Wallis H test was used to test the significant differences between the nest variables, like sizes of nests, nest entrance, and height of nests from water based on the nest-supporting plants by using Statistical Package for Social Science (SPSS version 25.0 software). Test of significance between the variables was assessed at $p<0.05$. The data sets did not satisfy the normality assumption and hence ANOVA test was not used. All the guidelines and protocols of nesting studies, as prescribed by Barve et al. (2020) were followed scrupulously. No eggs, chicks, juveniles, or adult birds were handled during the study. The locations of nesting sites were determined using a Garmin Etrex 20x GPS device. Photographs and videos were taken using a Nikon P1000 digital camera. The collected data were tabulated, analysed, and shown as graphical representation.

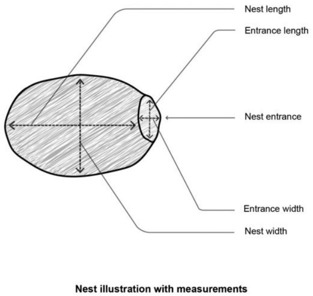


Fig. 2. Graphical representation of nest and mode of measurement of nest size.

Results

A total of 365 individuals of Tricoloured Munia and their 44 nests, on four nest-supporting plant species, were enumerated at five sites in the Veedur Wetlands. The maximum number of nests (29) were found on *Typha angustifolia*, and the minimum (1) on *Chrysopogon zizanioides* (Table 1). Paddy grew within c.50–100 m from all the nesting sites. A minimum distance of 0.25 km, and a maximum of 1.4 km was observed between adjacent nesting sites (Table 1). All nest-bearing plants were found partly submerged in the wetlands.

Table 1. Details of type nest-supporting plants, their life-forms, and number of nests enumerated in the study area

Site	Nest-supporting plant	Family	Life-form	Total number of nests enumerated	Percentage of nests (%)
Site-I	*Typha angustifolia*	Typhaceae	Herb	18	40.91
Site-II	*Typha angustifolia*	Typhaceae	Herb	11	25.00
Site-III	*Prosopis juliflora*	Fabaceae	Tree	10	22.73
Site-IV	*Vachellia nilotica*	Fabaceae	Tree	4	9.09
Site-V	*Chrysopogon zizanioides*	Poaceae	Herb	1	2.27
	Total			**44**	**100**

Nest construction

The birds had commenced nest construction during the third week of December 2021, and concluded breeding in the second week of March. Both the sexes were involved in nest construction. The birds had torn down dry pliant leaves of *T. angustifolia/Chrysopogon zizanioides*, one by one, in the vicinity, apart from bringing strands of fibre of *T. angustifolia* leaves far away from nesting sites, to make the skeleton for a nest. Then they added leaves, stems, and inflorescences of *Abutilon indicum*, and *Lantana camara, Cyanodon dactylon*, *Dactylis glomerata*, and *Saccharum spontaneum* to the skeleton, and finally constructed a ball-shaped nest with an entrance on the lateral side. In the case of trees, namely, *P. juliflora* and *V. nilotica*, they selected nest sites in dense canopies that were inaccessible to avian predators. Their nests comprised *T. angustifolia* and the other grasses mentioned above. The study revealed that the birds always moved in small flocks of three to five birds when bringing nesting materials from within a distance of *c*.100 m from the nesting sites. The birds brought feathers, and pieces of dried spongy spikelets of *S. spontaneum* and placed them in the egg lining of the nests. All these nests were constructed within the thick foliage of nest-supporting plants and were invisible from outside **[199. 3 a–e]**.

They constructed nests actively from 0630 to 1200 h daily, bringing in nesting materials; they were seldom observed ferrying nest materials after 1200 h. The study on 15 nests revealed that the birds took four to eight days to construct the nests.

The morphometric study on nests revealed that the length of nests varied from 18.5 cm to 21 cm, and width from 13.6 cm to 19.1cm (Table 2). The size of the entrance again varied: length 3.1cm to 4.4 cm, and width 2.9cm to 3.8 cm. The entrances of 79.5% nests (n=35) faced eastwards, whereas the entrances of 20.5% nests (n=9) faced westwards. No nest faced north or south.

Visuals and in-text indication

Visual items are intended to enrich the text content. The text guides how relevant the visuals are to the content.

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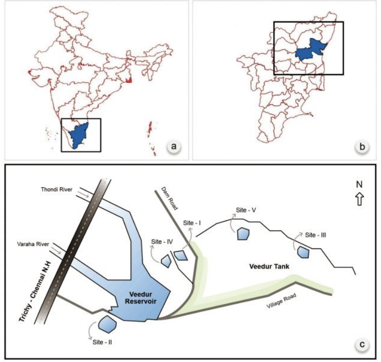

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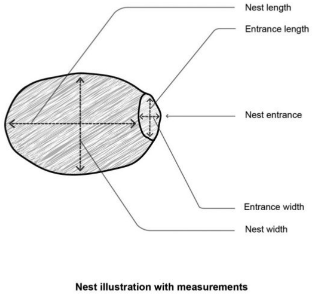

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In case you insert a photo

- Images, videos, and even graphics are intellectual property and must be cited on the slide, or in the speaker notes. Using photographs and videos from the internet without citing is illegal.
- For images, give credit to the photographer (usually listed in the caption).
- If you are unsure, use Pixabay, Unsplash, or the Noun Project (graphics) for free graphics and photographs.

Hatchling birds in the diet of the Blue Whistling Thrush *Myophonus caeruleus*

On 14 May 2022, at Barog (30.88°N, 77.07°E), District Solan, Himachal Pradesh, JSW observed a Blue Whistling Thrush *Myophonus caeruleus* killing the hatchling of a Streaked Laughingthrush *Trochalopteron lineatum* by thrashing it on a rock. The photographs **[210, 211]** show the hatchling being killed and held in the bill by the Blue Whistling Thrush. The action happened close to a busy trail. However, after killing the victim, the Blue Whistling Thrush flew away to a nearby perch, leaving the hatchling behind. It may have left because it felt that we were fairly close and it wanted us to leave. It might have come back later on but we had no way to confirm that as we left immediately after it flew away.

210. Blue Whistling Thrush killing the hatchling.

Both: J.S. Waraich

211. Blue Whistling Thrush holding the dead bird in the presence of an adult (parent?) Streaked Laughingthrush (foreground).

We had neither seen such behaviour earlier nor heard about any such incident from any other birders. This intrigued us to delve deeper and look for any reported/published observations about the Blue Whistling Thrush killing/feeding on small birds.

Citing Figures

When citing figures you did not create yourself, cite the document they appear in.

For example, if using this portion of a figure from the Augmented Vehicle Tracking paper above:

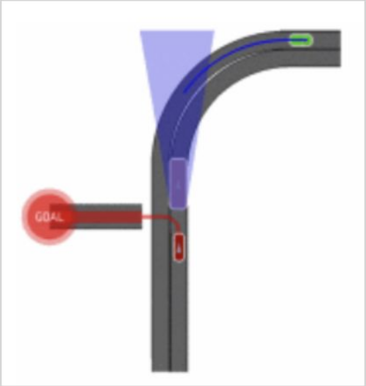

Fig. 3. Simulated intersection handling.
Source: Adapted from [2]

Then in your list of References, the article is listed normally in IEEE style.

[2] E. Galceran, E. Olson, and R. M. Eustice, "Augmented vehicle tracking under occlusions for decision-making in autonomous driving," in 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hamburg, Germany, 2015, pp. 3559–3565.

Class activity

Imagine you are interested in indoor air pollution by reading this article (https://www.nature.com/articles/d41586-023-00464-9)

- What graphs would be useful?
- Does the article show a reference for the further data?
- What information must be the most useful?
- What format (e.g., bar graph, pie chart, table) must be the most useful?
- How to cite the content?

Today’s learning objective- 2

Navigate the ways to evaluate common knowledge

What have we learned about plagiarism?

Definition?

Consequences?

Ways to avoid plagiarism?

Common knowledge?

How much do you remember about “common knowledge”?

Recap

Plagiarism exception: Common knowledge

The only source material that you can use in an essay without citing the source is material that is considered common knowledge.

This is because common knowledge is not attributable to one source.

Common knowledge refers to information that the average educated reader would accept as reliable without having to look it up.

Recap

What is common knowledge? What is not?

Information that most people know

Information shared by a cultural or national group

Knowledge shared by members of a certain field

Dataset generated by an individual or a group of people

Statistics obtained from sources such as any government website or international organizations

Studies done by others

Specific dates, numbers, or facts the readers would not know unless they had done the research.

Recap

How to determine common knowledge

To help you decide whether information can be considered common knowledge, ask yourself:

(1) Who is my audience?
(2) What can I assume they already know?
(3) Will I be asked where I obtained my information?

A description of the symptoms of Asperger's Syndrome would need to be cited for a composition in a general writing class but probably not need citation for an audience of graduate students in psychology.

A statement reporting that 24% of children under the age of 18 live in households headed by single mothers in a specific city needs to be cited. This information would not be known to the average reader, who may want to know where the figure was obtained.

Recap

Questions

Can this statement be considered as common knowledge?

The Big Bang theory posits that the universe began billions of years ago with an enormous explosion.

The phrase “Big Bang” was coined by Sir Fred Hoyle, an English astronomer. Hoyle used the term to mock the theory, which he disagreed with.

According to the Big Bang model, the initial explosion was produced when an infinitely hot, dense center referred to as a singularity, began to expand, giving rise to the particles that eventually formed into our universe.

Pair activity

Work with your partner and answer the questions in section 1 of the document.

https://docs.google.com/document/d/1MUdktamf7JtVplCo6dLKNkWibV8cEoCRTD9xh6IqFv0/

Debrief

Which question did you most struggle with?

In which of the situations presented did you feel like there was the most ambiguity?

Group activity

Work with your group members and answer the questions in section 2 of the document.

https://docs.google.com/document/d/1MUdktamf7JtVplCo6dLKNkWibV8cEoCRTD9xh6IqFv0/

Debrief

Which question did you most struggle with?

In which of the situations presented did you feel like there was the most ambiguity?